

Data Science

CSE558

Recap

- Chernoff's inequality

Overview

- Concentration Inequalities
- Universal Hashing

Chernoff Bound on Sum of Poisson Trials

For $\delta > 0$, setting $t = \ln(1+\delta) > 0$ we get,

$$\Pr(X \geq (1 + \delta)\mu) \leq \left(\frac{e^\delta}{(1 + \delta)^{(1+\delta)}} \right)^\mu$$

For $0 < \delta < 1$, setting $t = \ln(1-\delta) < 0$ we get,

$$\Pr(X \leq (1 - \delta)\mu) \leq \left(\frac{e^{-\delta}}{(1 - \delta)^{(1-\delta)}} \right)^\mu$$

Special Cases

Let $X_1, X_2, \dots, X_n$ be independent random variables such that for every $1 \leq i \leq n$,

$$\Pr(X_i = 1) = \Pr(X_i = -1) = \frac{1}{2} \quad \text{let, } X = \sum_{i=1}^n X_i$$

For any $a > 0$, prove that, $\Pr(|X| \geq a) \leq 2e^{-a^2/2n}$

$$\text{Proof: } \mathbf{E}[e^{tX_i}] = \frac{1}{2}e^t + \frac{1}{2}e^{-t} = \sum_{i \geq 0} \frac{t^{2i}}{(2i)!} \leq \sum_{i \geq 0} \frac{(t^2/2)^i}{i!} = e^{t^2/2}.$$

$$\mathbf{E}[e^{tX}] = \prod_{i=1}^n \mathbf{E}[e^{tX_i}] \leq e^{t^2 n/2}$$

Now, applying Markov's inequality,

$$\Pr(X \geq a) = \Pr(e^{tX} \geq e^{ta}) \leq \frac{\mathbf{E}[e^{tX}]}{e^{ta}} \leq e^{t^2 n/2 - ta}$$

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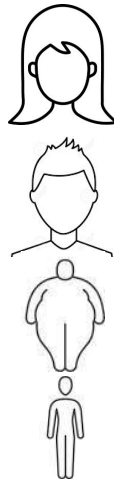
Proof: Applying Markov's inequality,

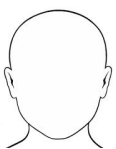
$$\Pr(X \geq a) = \Pr(e^{tX} \geq e^{ta}) \leq \frac{\mathbf{E}[e^{tX}]}{e^{ta}} \leq e^{t^2 n/2 - ta}$$

Setting, $t = a/n$ we get,

$$\Pr(X \geq a) \leq e^{-a^2/2n} \quad \text{and} \quad \Pr(X \leq -a) \leq e^{-a^2/2n}$$

Set Balancing



					—	
1	0	1	0	1	—	1
0	1	0	1	0	—	0
0	0	1	1	0	—	1
1	1	0	0	1	—	0

Dataset = A

#subjects = m

#features = n

Partition: $b \in \{-1, +1\}^m$

$$\min_{b \in \{-1, +1\}^m} \|Ab\|_{\infty}$$

Problem: Partition the subject into treatment group (T) and control group (C) such that each feature has a roughly equal number of subjects in each group.

Set Balancing - Analyze Solution

$$\min_{b \in \{-1, +1\}^m} \|Ab\|_\infty$$

Approximate solution: For $1 \leq i \leq n$, b_i is -1 or +1 with equal probability.

Let a_i be the i^{th} row of A having k 1's.

If $k \leq \sqrt{4m \ln n}$ then $|b^T a_i| \leq \sqrt{4m \ln n}$

If $k > \sqrt{4m \ln n}$ then consider the random variable $Z_i = b^T a_i$

Now applying Chernoff's inequality to bound Z_i ,

$$\Pr(|Z_i| > \sqrt{4m \ln n}) \leq 2e^{-4m \ln n / 2k} \leq \frac{2}{n^2}$$

Now taking a union bound, $\Pr\left(\|Ab\|_\infty \geq \sqrt{4m \ln(n)}\right) \leq \frac{2}{n}$

Special Cases

Hoeffding's Inequality: Let $X_1, X_2, \dots, X_n$, be n independent random variable such that $a_i \leq X_i \leq b_i$. Let their sum be $S = X_1 + X_2 + \dots + X_n$. Then for all $t > 0$,

$$\mathbf{P} (|S - \mathbf{E}[S]| \geq t) \leq 2 \cdot \exp \left(- \frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2} \right)$$

Bernstein's Inequality: Let $X_1, X_2, \dots, X_n$, be n independent random variable such that $|X_i - \mathbf{E}[X_i]| \leq b$. Let their sum be $S = X_1 + X_2 + \dots + X_n$ and $\sigma^2 = \mathbf{Var}[S]$. Then for all $t > 0$,

$$\mathbf{P} (|S - \mathbf{E}[S]| > t) \leq 2 \cdot \exp \left(- \frac{2t^2}{\sigma^2 + bt/3} \right)$$

Power of Chernoff

Bernstein's Inequality: Let $X_1, X_2, \dots, X_n$, be n independent random variable such that $|X_i - E[X_i]| \leq b$. Let their sum be $S = X_1 + X_2 + \dots + X_n$ and $\sigma^2 = \text{Var}[S]$. Then for all $t > 0$,

$$\mathbf{P} (|S - \mathbf{E}[S]| > t) \leq 2 \cdot \exp \left(-\frac{2t^2}{\sigma^2 + bt/3} \right)$$

Compare Chebyshev and Chernoff when, $t = k\sigma$ and $b \leq \sigma$.

Overview

- Concentration Inequalities
- Universal Hashing

Hashing



Query: Bought ticket?



Linear search: Time consuming. **Can we do better?**

Hash Table

U: Universe (All people who have pre-registered for the event. Typically its large.)

n: number of tickets sold (Set of people who bought tickets. Typically $n < |U|$.)

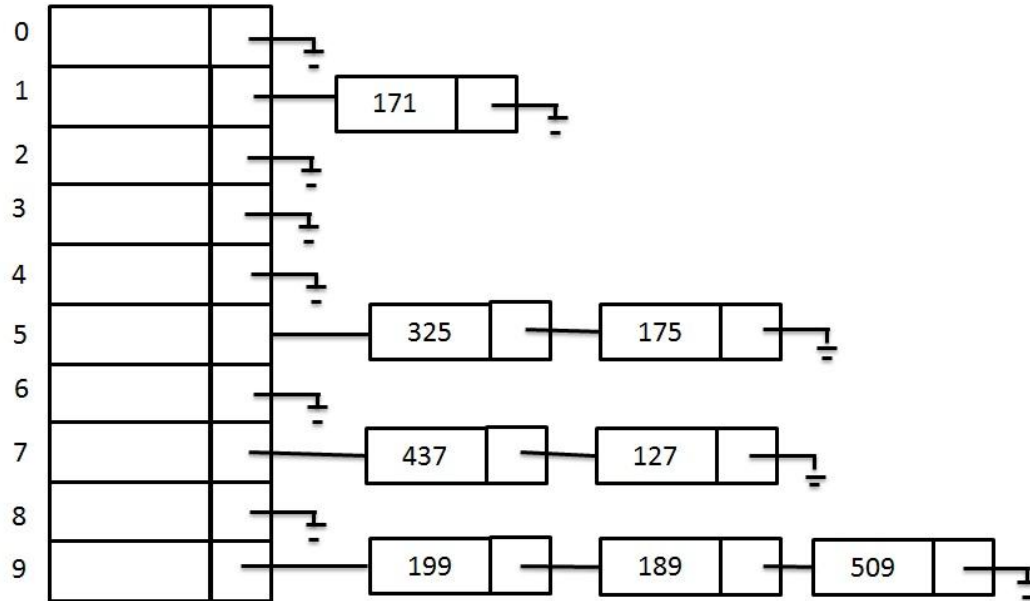
Hash Table

- Array of size m
- Hash function, $h: U \rightarrow \{0, 1, \dots, m-1\}$

Typically, $m \ll |U|$ and $m < n$.

- Collision happens when $x \neq y$ but $h(x) = h(y)$.
- By pigeonhole principle.

Handle Collision



At every index, maintain a chain of all the elements hashed there using a linked list.

What is a good hash function?

Hash function

- Deterministic hash function is not good enough.
 - $h(x) = x \bmod m$.
 - Too many collision if adversary knows the hash function.
 - Leads to linear search in worst case scenario.
- Random hash function
 - $H = \{h_1, h_2, \dots\}$
 - Choose $h \in H$ uniformly at random.
 - Analyze expected query time.
- $|H|$ cannot be too large, e.g., $m^{|U|}$
 - Defeats the purpose quick membership check!

Hash Family

Create a small hash family with good behaviour.

- Universal: $\Pr_{h \in \mathcal{H}} (h(x) = h(y)) = \frac{1}{m}$, for every x and y